

Institute of Actuaries of India

Subject SA6 –Investment

September 2018 Examination

INDICATIVE SOLUTION

Introduction

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable.

Solution 1:

i)

a) **Cost:**

- There is an extra layer of cost in outsourcing in the margin to the insurer.
- There is a cost saving in that the fund does not need actuarial valuations, audit reports anymore.

b) **Flexibility / control over bonus declarations / increases:**

- When transferred the increases will be according to a formula
- The trustees and employer have no control over these increases
- Cannot adjust levels of pension with ad hoc increases

c) **Control over the assets:**

- If fund stays on balance sheet, trustees and employer have control over asset manager selection, asset allocation, etc.
- If transferred, assets belong to insurer who invests according to its risk profile
- A mandate between the fund and insurer could specify some guideline as to how assets could be invested and with whom.

d) **Relationship with the pensioners:**

- If transferred trustees and employer lose this relationship, which might be an issue for some pensioners who wants to look to the employer to look after them in retirement
- Insurer takes over full responsibility and reputational risk

e) **Mortality / longevity risk:**

- Insurer carries this risk in transferred model that pensioners live longer than assumed in valuations
- Part of the added cost of outsourcing as insurer needs to cost for this risk
- Big part of the benefit to employer to be released of this risk
- On a with-profit type product / basis of outsourcing some of this risk is shared by the pool of participants as future increases may be adjusted to allow for this.
- Transferred in non-profit product leaves all the risk with insurer vs on balance sheet where increases are always a target and not fixed, i.e. some offset for mortality allowed
- This gives pensioners more certainty

f) Administration:

- If transferred the employer and fund does not have to do any admin again, i.e. paying members, reconciling payments, tax certificates, etc.
- Even though the insurer will cost for it, it might still be a cost saving as well, because of scale benefits
- Admin around proof of life, i.e. only paying pensioners that are still alive falls in insurer, and benefits pensioners if done effectively – reduction in risk for employer

g) Size of the fund / employer:

- A big fund / large employer could take some of the risks on balance sheet
- Less chance for a small fund to get suitably diversified assets
- And bigger effect of mortality
- Prohibits investment in items like property
- Size of pensioner liability vs company balance sheet also important – if smaller percentage company may be less worried about out-sourcing

h) Investment risk:

- Removed from employer and fund if transferred
- Types of investments available through scale in insurer much wider than fund alone
- Assets in insurer partially constrained because of risk of matching, capital and other regulatory constraints vs own fund
- Less so for DB funds as employer balance sheet backs liabilities
- Derivatives and other sophisticated non-standard assets can be invested in by an insurer with more expertise and balance sheet backing it
- What are current market return expectations and to what extent (given cost) can hedging be done within fund?
- Are there currently any illiquid assets?
- Funding level of scheme?

i) Trustee involvement:

- No need if transferred
- If wanted, keep on balance sheet

j) Employer:

- Risk totally removed if transferred
- Comment on accounting liabilities versus buyout, cost of capital, VAR

impact on balance sheet and P/L and cash contributions.

k) **Price of transfer:**

- Margin to insurer
- Assumptions used in calculations
- Capital cost of guarantees
- Fund/employer might not have enough money to transfer
- Surplus exercise done, finalized
- Core business of employer - Pensions is not part of core business of the employer, but forms big part of balance sheet liabilities if kept
- Understanding the pensioner liability
- Need to differentiate between contractual liability and PRE
- Is it a closed group?
- Are there different categories of pensioners in which case only some may be transferred?

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ii)

a) **Cash flow matching principles**

- invest assets to match the liability
- liability is the profile and incidence of pension payments i.e. is not the same as immunization/duration matching
- assets needs to have similar cash flows and behave the same in different market movements
- which implies the $PV(A) = PV(L)$
- in stressing the interest rates all along the curve for parallel and nonparallel movements for both assets and liabilities
- Liability is series of guaranteed existing pensions in payment long into the future.
- Increases with CPI
- Will decrease with members dying or inflators less than demographic/economic assumptions
- Assets invested to deliver projected cash flows
- Similar to incidence of pension payments through coupon payments and bond maturities
- Bonds managed in right maturities and exposure to achieve the cash flows desired
- Can also be done by using swaps to swap existing cash flows for the desired ones

b) **Devising a strategy**

Assumptions

- Payments happen monthly
- Increases annually
- Decrements happen continuously
- Government bonds and other high quality liquid bonds have varying maturities not covering every cash flow
- Coupons get paid every 6 months

Methods and considerations

- For cashflow matching - one can group a series of payments together and match those with the appropriate cash flows generated using the above assumptions.
- With swaps you can match more closely – government bonds may not exist or have required cap/collar features. You can construct the term and size of each payment through swaps. One can use a combination of bonds and swaps.
- Counter-party risk different between bonds and swaps
- If matching with bonds the fund will always be fully exposed. With swaps it may vary over life-time of swap.
- Swap yields differ to bond yields. Although highly correlated, they often move out of line from each other, because of liquidity, bank vs government risk, demand vs supply.
- If the liabilities are valued off the bond curve, you will run a basis risk if the actual assets are invested in swaps and vice versa.

Swaps have the following advantages above bonds:

- More accurate matching
- Swaps allow greater flexibility
- You do not have to lay out cash for the exposure, i.e. it is effectively an overlay derivative transaction
- Cash can therefore be invested to deliver higher returns, while the desired cash flows are matched by using swaps
- You can swap fixed cash flows for floating cash flows as well
- The cash returns can be enhanced by adding credit assets to the underlying investment instead of the pure cash
- If swap yields are below that of bond yields it would mean that the return from swaps will not be adequate to cover the liabilities or rather that bonds are cheaper.

Additional Considerations :

- The scheme has equity linked type increases
- With an underlying guarantee that pensions cannot reduce
- Valued using stochastic modelling, sophisticated methods to reproduce the payoff closely
- The trick is to find the right balance of cash flow matching and equity type investments

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iii) Full versus partial matching

<i>Match guaranteed CFs</i>	<i>Match guaranteed CFs and inflationary increases</i>	<i>Partial Matching</i>
Less interest rate exposure as most part hedged	Very low risk option for employer	low risk option for employer
Low risk option for employer	Will be the most expensive option, so might need additional funds at inception	reinvestment risk exists if only the early part of liability hedged
Will need a big part of assets to match the guaranteed liability	No upside for pensioners as inflation increases effectively guaranteed through matching strategy.	some interest rate exposure will exist and make assets relatively volatile against liabilities
Which leaves little left to provide growth for future increases	Matching might not be perfect as very few inflation linked bonds in issue at all durations, especially ultra long.	more assets left over for growth and future increases
Very little link to inflation so increased inflation risk	Could be the most desirable outcome for trustees, shareholders and remaining members as risk of inadequate assets is lowest.	This is better than choosing a high risk option.
Biggest risk is matched so can be quite aggressive with remaining assets to provide growth/increases.		choice of remaining assets critical to provide growth
Actual mortality experience and other movements will impact the future matching program, which does leave some reinvestment risk		link to inflation not clear and direct, will depend on growth assets, some risk exist

[10]

iv) Expected inflation is priced within the nominal rates :

Nominal Rates = Risk Free rates plus Expected Inflation.

Therefore one can easily hedge expected inflation by investing in bonds/ other instruments which deliver yields at least equal to nominal rates.

It is unexpected inflation that is difficult to hedge and poses issues.

[2]

- v) The risk of the impact of high inflation on the Company's retirement benefit schemes and their sponsors, and on an individual's pension purchasing power, must be weighed against the costs of protecting against such a scenario.

Conversely, are there risks in the portfolio if inflation and inflation expectations decrease?

Assets

- Levels of inflation different to the "norm" or different to what is expected can have an impact on the value of assets. For example: the effect of deflation in Japan on Japanese equities.
- The effect on equities in emerging markets of high growth and inflation, and how this might change.
- The impact of "cooling" measures which can affect asset prices.
- Increasing inflation makes fixed income assets increasingly unattractive and can reduce prices.
- Certain asset values, such as for commodities, are directly linked to (or indeed a cause of) increases in inflation.
- Different asset classes perform differently in different inflationary environments – being in the right asset class at the right time is key.
- Any reduction in the value of assets impacts the deficits within the Company's defined benefit scheme. In a defined contribution scheme, it would lead to lesser benefits than envisaged for members and hence it could prove to be a source of concern for the employees covered by the defined contribution plan.

Liabilities

- Many of the benefits in final salary pension schemes are linked to inflation:
- The link between salary growth and inflation.
- The increase to deferred members' benefits.
- Increases to pensions already in payment.
- Higher inflation therefore increases the benefits that need to be paid out.
- Higher inflation expectations increases the value placed on the liability cashflows and has funding impacts for the scheme and sponsor.
- Within a defined benefit scheme, inflation can adversely impact the final cost to the employer and hence increase the deficits within the scheme.

Defined Contribution/Money Purchase Schemes

- It is expected that Trustees and employers provide adequate fund options for defined contribution/money purchase scheme members to be able to mitigate the risks of inflation.
- Default options in these schemes should take consideration of the need for members' retirement pots to grow in real terms (i.e. after inflation) over the period of their working lives.

vi) Issues to consider when hedging inflation

When thinking about hedging, the following factors need to be considered:

- a. What are your risk/reward tolerances?
- b. How much inflation sensitivity is currently in the pension scheme assets and liabilities?
- c. What are the trustees' and markets' expectations regarding future inflation levels?
- d. What assets are available to hedge inflation and what is their current price?
- e. Are there any liability management exercises planned for the pension scheme?
- f. Are there any other issues specific to your scheme, which need to be considered?

Alternative mechanism to hedge inflation

Asset	Pros	Cons
Inflation swap	▪ Targets only inflation, ▪ Unfunded	▪ Introduces basis risk compared to funding basis, ▪ Transaction cost could be higher than inflation-linked govt bonds (especially at the longer end of the curve).
Inflation-linked gilt on asset-swap	▪ Targets only inflation ▪ Better inflation rate than swap	▪ Funded (although could use repos), ▪ New type of derivative (selling duration)
Total Return Swap on index-linked gilts	▪ Unfunded ▪ Could also hedge out nominal risk	▪ New type of derivative
Long-dated property leases	▪ Collateralized by property	▪ Not a mark-to-market hedge, ▪ Property risks, ▪ Lead time, ▪ Illiquid
Corporate linkers/corporate inflation linked note	▪ Additional credit spread ▪ Mark-to-market hedge	▪ Limited market, ▪ Counterparty risk, ▪ Lead time
Established infrastructure	▪ Operational infrastructure can be lower risk ▪ Can be government	▪ Not a mark-to-market, ▪ Counterparty risk,

	backed/funded	▪ Lead time, ▪ Illiquid
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[15]

[70 Marks]

Solution 2:

i)

- **Simplification, less confusion :-** The market regulator sought to bring uniformity in the characteristics of similar schemes to ensure that an investor is able to evaluate various options before making any investment.
- **Better comparison:-** Earlier, in the absence of any standard definition of MF categories, mutual fund companies and research & advisory firms came up with their own methodologies of categorising funds. This made it difficult for investors and advisors to evaluate and compare funds and select suitable ones for investment. Due to standard categorization, comparison of schemes will be easier and one would be able to conduct an apple-to-apple comparison for each category. A better comparison would help in decision making. [1]

ii)

Benefits

- Apart from the ones mentioned above Merging of various schemes might bring uniformity in commission paid by asset **management companies (AMCs)**
- **Scheme merger will bring down the number of portfolios to be managed, thereby giving time to fund managers to focus their efforts on generating alpha**

Issues

- **The issue this creates for existing investors is that they have no say in any of these changes. The only option they have is to stick to the current scheme or switch. The shift may also be needed to retain their debt / equity allocation or to maintain the segment wise allocation.**
- **They may need to bear tax issues arising out of exits.** [2]

iii)

- Revisit your portfolio. You will have to take a fresh look at your portfolio. While you will have to revisit the funds in your portfolio. it shouldn't be done in haste.
- Check whether the change is cosmetic or goes much deeper. Funds have to offer names that reflect the true nature of the portfolio. Words like "opportunities", "advantage" and "prudence" could be misleading. So many schemes will undergo a change in names.
- There will be scheme mergers. The AMC will try their best to ensure that a union of schemes ensures finding suitable matches. But investors must do the same due diligence.
- Take a holistic view. Even post a change in mandate, you may still approve of the scheme. But view it now in conjunction with your entire portfolio. For instance, if you had just one large-cap fund in your portfolio and it has been reclassified as a multi-cap fund, you may want to see how it fits in with your other investments

- Keep the tax impact in mind. Make personalized calls and avoiding blanket opinions. The review has to be done based on risk profile and the fund's new avatar.
- Don't view any fund in isolation. Make every decision keeping the rest of your portfolio in mind

The challenges that needs focus are

- bear the burden of capital gains tax when you opt to exit from schemes in case of mergers.
- Fund managers may have to reshuffle scheme portfolios every six months, which will increase their costs and impact their returns.
- The nature of risk carried by debt schemes may still not be understood well by investors as simply changing the name might not highlight the quantum of risk element in these schemes.
- Hybrid schemes, under the current categorisation, have been defined. However, for an investor, the scheme differentiator will still remain a concern.

[5]

iv)

- Churn in portfolio: Every six months, fund managers may be required to reshuffle portfolios based on investment categorisation (large, mid and small) published by the Association of Mutual Funds in India (AMFI). This may lead to higher portfolio turnover and an increase in transaction costs, directly impacting fund returns.
- Risk of front-running: Re-categorisation of certain stocks from large to mid, mid to small or vice versa on a half-yearly basis may result in the front-running of these stocks.
- Increased cost due to scheme mergers: The merger of schemes will result in the renegotiation of distributor commissions and the management of trails, thereby increasing transaction costs.
- Shrinking of the mid-cap universe: Post SEBI's regulation, there will be only 150 companies categorised under the mid-cap universe as compared to the current 400 stocks, as a result of which fund managers will have limited options to invest under the mid-cap category.
- Full market capitalisation instead of free float for ranking: The new guideline categorises stocks based on full market cap instead of free float. This will create liquidity impact among stocks. Certain stocks which may rank high in case of free float might rank low under the full market cap method

[4]

[12 Marks]

Solution 3:

i) Key conditions applicable to PEFs investing in Indian insurance companies as a "Promoter"

- A PEF can invest in an Indian insurance company as a "promoter" only through an SPV and not directly;
- A PEF through an SPV shall not be a promoter for more than one life insurer, one general insurer, one health insurer and one reinsurer;
- Lock-in period of five years for the (i) SPV; and (ii) shareholders of the SPV holding atleast 10% capital of the SPV;

- Any new shareholders in the SPV through issue of fresh shares beyond 25% would require prior IRDAI approval;
- Minimum shareholding by the promoters/ promoter group to be 50% of the paid up equity share capital of the insurance company at all times. However, if the present holding by the promoters is below 50%, such lower holding would be the minimum holding;
- The Indian insurance company to comply with IRDAI guidelines on "Indian owned and controlled" and Indian Insurance Companies (Foreign Investment) Rules, 2015;
- Schemes to be filed with SEBI in accordance with the provisions of relevant SEBI regulations;
- Investment to be as per the PEF's strategy reflected in its placement memorandum;
- Investment to be made entirely out of own funds and not borrowed funds;
- Investment memorandum or charter documents of the investor/ investing vehicle to permit investment up to the permitted limits including any future capital requirements;
- Investment to be subject to the prescribed "Fit and Proper" criteria. One of the key criteria is "whether the applicant is a widely held entity, publicly listed and a well established regulated financial entity in good standing in the financial community";
- PEFs to provide a specific undertaking that it will not create any encumbrance on or leverage the investment;
- Either the Chairman of the Indian insurance company to be an independent director or its Chief Executive Officer/ Managing Director/ Whole-time director to be a professional and not a nominee of a promoter;
- Atleast one-third of the board of the insurance company to comprise of independent directors;
- An undertaking to subscribe to the rights issue of the Indian insurance company to be provided to ensure that the Indian insurance company is not cash strapped;
- An undertaking of the post lock in period divestment plan (preferably through an Initial Public Offer) in accordance with the relevant regulations to be provided

[6]

ii) Key conditions applicable to PEFs investing in Indian insurance companies as an "investor"

- A PEF may invest in an Indian insurance company as an "investor" directly or through a Special Purpose Vehicle (SPV) i.e. an onshore company or limited liability partnership incorporated in India;
- The PEF shall hold maximum 10% of the paid up equity share capital of the insurance company;
- All Indian investors along with the PEFs to jointly hold maximum 25% of the paid-up equity share capital of the insurance company;

[2]

[8 Marks]

Solution 4:

i)

a) What are the details of the foundation?

- Size of fund
- Funding position
- Future financial commitments and income
- Appetite for risk

- Regulatory Permitted assets
- Socially responsible, sustainable or ethical investment policies
- Existing asset management arrangements
- Governance and administration structure
- Justification for existing investment policy

- b) How would the exposure to commodities be achieved? [reference to real estate exposure may give clues to size/management approach]
- c) Would holdings be restricted to long only? Or should short positions be considered?
- d) Which commodities? Energy/precious metals/base metals/agricultural/meat and livestock

[4]

ii)

- **Nature:** real / fixed - commodities can be argued to perform well in inflationary environments hence provide some inflation matching although they are neither fixed or real in nature.
- **Term** — can be argued that commodities are short term, not appropriate for matching longer dated liabilities.
- **Currency** — commodity prices commonly in US dollars, liability INR — would require currency swap, introducing further element of cost.
- **Certainty** — commodity prices can be volatile in both the short and longer term. There is no direct link between their price movement and the valuation of liabilities. This will introduce volatility into the funding position.
- Commodity holdings do not produce income and are therefore not
 - suitable for “matching” future cashflow commitments.
- It is possible to make significant profits from investing in commodities in short periods, however there is also the opportunity to lose large amounts in equally short periods.
- Diversification — commodities offer significant real returns that are produced by doing real economic work within the economy.
- The returns accrue to long only investors without the need for *active* management but there will probably remain the need for third party management in terms of selection, market access, trading.
- Recommended management approach and costs.
- Reasoned argument for appropriate size of allocation.

[6]

[10 Marks]
